

--	--	--	--	--	--	--	--

B.Tech. Degree V Semester Supplementary Examination June 2024

ME 19-205-0502 MECHANICS OF MACHINERY (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Understand the principles of kinematic pairs, chains and their classification, DOF, inversions, equivalent chains and planar mechanisms.

CO2: Analyze the planar mechanisms for position, velocity and acceleration.

CO3: Synthesize planar four bar and slider crank mechanisms for specified kinematic conditions.

CO4: Evaluate gear tooth geometry and select appropriate gears for the required applications.

CO5: Design cams and followers for specified motion profiles.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

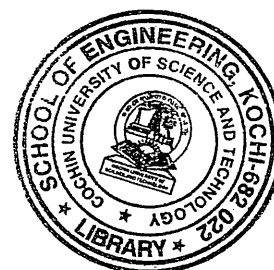
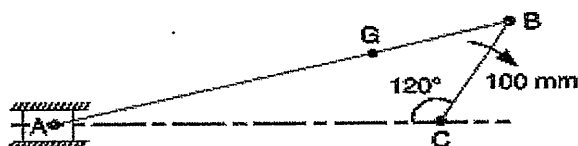
(Answer ALL questions)

		(8 × 3 = 24)	Marks	BL	CO	PO
I.	(a)	Define degree of freedom, what is meant by mobility of a mechanism?	3	L1	1	1
	(b)	A pin joins two links A and B. A rotates with ω_A angular velocity and B rotates with ω_B angular velocity in opposite direction. What is the rubbing velocity of that point?	3	L3	1	1
	(c)	What is expression for coriolis component of acceleration? When does this component come into picture?	3	L3	2	1,2
	(d)	What is meant by pressure angle with respect to Cam Dynamics?	3	L2	2	1,2
	(e)	State the Law of Gearing. What is its significance?	3	L3	3	1,2,3
	(f)	What is the peculiarity of an epi-cyclic gear train? Give its application.	3	L2	3	1,2,3
	(g)	Define the angle of repose.	3	L2	4	1,2,3
	(h)	What is a dynamometer? Name any two types of Dynamometers.	3	L1	4	1,2,3

PART B

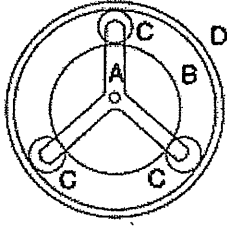
(4 × 12 = 48)

- II. Explain with help of diagrams, inversions of a slider crank chain. 12 L1 1 1
- OR
- III. An engine mechanism is shown in the figure. The crank CB = 100 mm and the connecting rod BA = 300 mm with centre of gravity G, 100 mm from B. In the position shown, the crankshaft has a speed of 75 rad/s and an angular acceleration of 1200 rad/s². Find: 12 L3 1 1
- (i) velocity of G and angular velocity of AB.
- (ii) acceleration of G and angular acceleration of AB.



(P.T.O.)

BTS-V(S)-06-24-2688

		Marks	BL	CO	PO
IV.	A pinion having 30 teeth drives a gear having 80 teeth. The profile of the gears is involute with 20° pressure angle, 12 mm module and 10 mm addendum. Find the length of path of contact, arc of contact and the contact ratio.	12	L2	2	1,2
OR					
V.	In an epicyclic gear of the 'sun and planet' type shown in the figure, the pitch circle diameter of the internally toothed ring is to be 224 mm and the module 4 mm. When the ring D is stationary, the spider A, which carries three planet wheels C of equal size, is to make one revolution in the same sense as the sun wheel B for every five revolutions of the driving spindle carrying the sun wheel B. Determine suitable numbers of teeth for all the wheels.	12	L3	2	1,2
					
VI.	A cam drives a flat reciprocating follower in the following manner: During first 120° rotation of the cam, follower moves outwards through a distance of 20 mm with simple harmonic motion. The follower dwells during next 30° of cam rotation. During next 120° of cam rotation, the follower moves inwards with simple harmonic motion. The follower dwells for the next 90° of cam rotation. The minimum radius of the cam is 25 mm. Draw the profile of the cam.	12	L2	3	1,2,3
OR					
VII.	The mean diameter of the screw jack having pitch of 10 mm is 50 mm. A load of 20 kN is lifted through a distance of 170 mm. Find the work done in lifting the load and efficiency of the screw jack when (i) the load rotates with the screw. (ii) the load rests on the loose head which does not rotate with the screw. The external and internal diameter of the bearing surface of the loose head are 60 mm and 10 mm respectively. The coefficient of friction for the screw as well as the bearing surface may be taken as 0.08.	12	L2	3	1,2,3
VIII.	A plate clutch has three discs on the driving shaft and two discs on the driven shaft, providing four pairs of contact surfaces. The outside diameter of the contact surfaces is 240 mm and inside diameter 120 mm. Assuming uniform pressure and $\mu = 0.3$; find the total spring load pressing the plates together to transmit 25 kW at 1575 r.p.m. If there are 6 springs each of stiffness 13 kN/m and each of the contact surfaces has worn away by 1.25 mm, find the maximum power that can be transmitted, assuming uniform wear.	12	L3	4	1,2,3
OR					
IX.	Derive Freudenstein's equation for a four-bar mechanism.	12	L2	4	1,2,3

Bloom's Taxonomy Levels

L1 – 15%, L2 – 47.5%, L3 – 37.5%.
